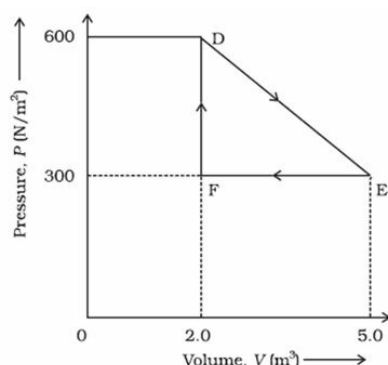


CBSE Test Paper 02
Chapter 12 Thermodynamics

1. A thermodynamic system is taken from an original state to an intermediate state by the linear process shown in Figure. Its volume is then reduced to the original value from E to F by an isobaric process. Calculate the total work done by the gas from D to E to F **1**

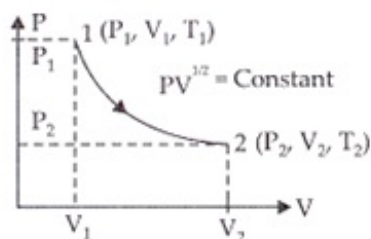


- a. 500 J
b. 480 J
c. 450 J
d. 470 J
2. A 1.0-kg bar of copper is heated at atmospheric pressure. If its temperature increases from 20° C to 50° C, what is the work done by the copper on the surrounding atmosphere? Given that for Cu $3\alpha = 5.1 \times 10^{-5} / ^\circ\text{C}$, $\rho_{Cu} = 8.92 \times 10^3 \text{ Kg/m}^3$ **1**
- a. 1.7×10^{-2} J
b. 1.8×10^{-2} J
c. 1.6×10^{-2} J
d. 1.9×10^{-2} J
3. In Thermodynamics Extensive variables depend **1**
- a. on the 'temperature' of the system
b. on the 'size' of the system
c. on the 'entropy' of the system

d. on the 'pressure' of the system

4. An ideal gas is enclosed in a cylinder with a movable piston on top. The piston has a mass of 8 000 g and an area of 5.00 cm^2 and is free to slide up and down, keeping the pressure of the gas constant. How much work is done as the temperature of 0.200 mol of the gas is raised from 20.0°C to 300°C ? **1**
- a. 456 J
b. 486 J
c. 476 J
d. 466 J
5. A steam engine delivers $5.4 \times 10^8 \text{ J}$ of work per minute and services $3.6 \times 10^9 \text{ J}$ of heat per minute from its boiler. What is the efficiency of the engine? How much heat is wasted per minute? **1**
- a. 14 percent, $3.0 \times 10^9 \text{ J}$
b. 12 percent, $2.8 \times 10^9 \text{ J}$
c. 13 percent, $2.9 \times 10^9 \text{ J}$
d. 15 percent, $3.06 \times 10^9 \text{ J}$
6. What is the efficiency of Carnot engine Working between ice point and steam point? **1**
7. If the temperature of the sink is increased, what will happen to the efficiency of Carnot engine? **1**
8. A cylinder containing one gram molecule of the gas was compressed adiabatically until its temperature lose from 27°C to 97°C . Calculate the work done and heat produced in the gas. Take γ as 1.5. **1**
9. Calculate the fall in temperature when a gas initially at 72°C is expanded suddenly to eight times its original volume. ($\gamma = 5.3$) **2**
10. State Kelvin-Planck statement of Second Law of Thermodynamics. What is the significance of this law? **2**
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11. State Carnot's theorem. What is its significance? **2**
12. At 0°C and normal atmospheric pressure, the volume of 1g of water increases from 1cm^3 to 1.091 cm^3 on freezing. What will be the change in its internal energy? Normal atmospheric pressure is $1.013 \times 10^5 \text{ N/m}^2$ and the latent heat of melting of ice is 80 cal/g . **3**
13. A perfect Carnot engine utilizes an ideal gas the source temperature is 500 K and sink temperature is 375 K . If the engine takes 600 kcal per cycle from the Source, Calculate **3**
- The efficiency of engine
 - Work done per cycle
 - Heat rejected to sink per cycle
14. Two Carnot engines A and B are operated in series. The first one A receives heat at 900 K and rejects it to a reservoir at temperature T . The second engine B operates on this reservoir and rejects heat to a reservoir at 400 K . Calculate temperature T when **3**
- efficiencies of both A and B are equal.
 - the work outputs of both A and B are equal.
15. Consider a P-V diagram in which the path followed by one mole of perfect gas in a cylindrical container is shown in Figure. **5**



- Find the work done when the gas is taken from state 1 to state 2.
- What is the ratio of temperature T_1/T_2 , if $V_2 = 2V_1$?
- Given the internal energy for one mole of gas at temperature T is $(\frac{3}{2}) RT$, find the heat supplied to the gas when it is taken from state (1) to (2) with $V_2 = 2V_1$.